ZIXIAO WANG

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EDUCATION

Peking University Beijing, China
Bachelor of Artificial Intelligence Aug. 2023 – Jul. 2027(expected)

- The Zhi Class (honors program) of 2023 at the department of EECS
- Overall GPA: 3.64/4

Carnegie Mellon University Pittsburgh, United States
Exchange student in School of Computer Science Jan. 2026 – May. 2026

University of Wisconsin Madison Madison, United States
Summertime research intern May. 2026 – present

HONORS

- Liao Kaiyuan Scholarship of Peking University, 2023–2024 academic year
- Merit Student of Peking University, 2023–2024 academic year
- Peking University Academic outstanding award, 2023–2024 academic year

RESEARCH INTERESTS

Large Language Models; Optimization; Pretraining; Efficient Reinforcement Learning; Multimodal

PUBLICATIONS AND MANUSCRIPTS

(* denotes equal contribution.)

- Zixiao Wang*, Yifei Shen*, Huishuai Zhang.
  OLion: Approaching the Hadamard Ideal by Intersecting Spectral and ℓ_∞ Implicit Biases.
  Accepted by *ICML 2026*. [ICML poster] [arXiv]
- Zixiao Wang, Shi-Yuk Fong, Jiawei Zhang.
  ITO-HC: Identity-Centered Tanh Off-Diagonal Hyper-Connections.
  Submitted to *ICLR 2027*. [paper]
- Jiayi Gao, Minghang Zheng, Zixiao Wang, Yujie Zhong, Zilong Zheng, Yang Liu.
  Training-free Video Temporal Grounding Enhanced by Contextual Region Cue based Test-time Adaptation.
  Submitted to *AAAI 2026*. [paper]

RESEARCH EXPERIENCE

OLion: Approaching the Hadamard Ideal by Intersecting Spectral and ℓ_∞ Implicit Biases June 2025 – Jan. 2026
First author: Zixiao Wang*
State Key Laboratory of General Artificial Intelligence, Peking University
Advisors: Prof. Huishuai Zhang and Prof. Dongyan Zhao

- Developed OLion, combining Muon-style orthogonalization and Lion-style sign updates to control spectral and coordinate-wise ℓ_∞ scales.
- Analyzed convergence under diagonal isotropy; implemented the optimizer and led experiments and paper writing for GPT-2/Llama pretraining, SiT pretraining, and supervised fine-tuning.

ITO-HC: Identity-Centered Tanh Off-Diagonal Hyper-Connections May. 2026 – Oct. 2026

First author: Zixiao Wang
University of Wisconsin–Madison
Advisor: Prof. Jiawei Zhang (University of Wisconsin, Madison)

- Designed ITO-HC, an identity-centered residual mixer that preserves an exact unit diagonal and adds bounded, signed off-diagonal corrections. It retains dynamic pre/post routers while removing the dynamic residual-mixer generator.
- Analyzed layer-wise routing and derived conditional singular-value stability bounds. Swept GPT-2-style models from 4 to 64 streams on OpenWebText, with baseline comparisons at four streams.

Training-free Video Temporal Grounding Enhanced by Contextual Region Cue based Test-time Adaptation Nov. 2024 – Mar. 2025
Coauthor; contributed to code and paper writing
Peking University
Advisor: Prof. Yang Liu (Peking University)

- Built a training-free video-level adapter using BLIP-2 Q-Former region tokens. A Positive Cache boosts low-quality frames within an event; an Outlier Cache uses temporal clustering to identify and suppress visually similar but irrelevant frames.
- Integrated the cache-based region scores with TFVTG localization while keeping the VLM frozen. The paper reports gains over TFVTG on Charades-STA, ActivityNet-Captions, QVHighlights, and several out-of-distribution settings.

Zero-order Reinforcement Learning Jul. 2026 – present
University of Wisconsin–Madison
Advisor: Prof. Jiawei Zhang

- Ongoing project investigating zero-order optimization methods for reinforcement learning when gradient information is unavailable or expensive.

ACADEMIC PROJECTS

Final project for Computer Vision 2024

- Generate illusion images using diffusion, enabling text and image input as controlling context
- Open source: [GitHub]

Final project for Introduction to Visual Computing and Interaction 2024

- Apply path tracing algorithm on the rendering of Cornell Box scene
- Open source: [GitHub]

COMMUNITY SERVICE

- Reviewer, *ICLR 2027*

SKILLS

Programming: C++, C, Python (PyTorch, TensorFlow), Mathematica, LaTeX
Tools: vLLM, DeepSpeed, Hugging Face, WandB, Docker, Git
English: TOEFL: 105 ; CET-6: 668